

One fixed satellite family almost norms every detected unit vector

MathlibAnnex.Satellite.absoluteMaximizer_satellites_almost_norm

theorem

Uses a detecting coefficient and its matching satellite without replacing the family.

Statement

Let $n \in \mathbb{N}$ and use reference coordinates $V = \mathbb{R}^n$ with the sup norm. Let $p : V \rightarrow \mathbb{R}$ be a continuous seminorm with specified constants $a_p, b_p > 0$ such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in V).$$

Thus p is a norm. We use p for the model distance and $\|\cdot\|_\infty$ for reference-coordinate estimates; the two norms are allowed to coincide. A p -dual contraction is a continuous real linear functional r satisfying $|r(x)| \leq p(x)$ for every x . Let $e_1, \dots, e_n$ be the standard coordinate vectors. For an ordered family $F = (\ell_i)_{i=1}^n$ of continuous real linear functionals on V , put

$$M_F = (\ell_i(e_j))_{i,j}, \quad T_F x = (\ell_i(x))_i = M_F x, \quad \mathcal{U}_p = \{F : \forall i, x, |\ell_i(x)| \leq p(x)\}.$$

For a real number η , define the attained maximum and near-maximal set by

$$D_p = \max_{F \in \mathcal{U}_p} |\det M_F| > 0, \quad \mathcal{N}_{p,\eta} = \{F \in \mathcal{U}_p : D_p - \eta \leq |\det M_F|\}.$$

Admissibility is measured with p ; the matrices are written in the fixed reference coordinates.

Let J be a finite index set, possibly empty; choose coefficients $c_a \in V$ for $a \in J$. A configuration is a pair $Q = (F, s)$, where F is a frame and $s = (s_a)_{a \in J}$ is a family of additional continuous real linear functionals, called satellites. Write

$$\Omega_p^J = \{(F, s) : F \in \mathcal{U}_p, \forall a, x, |s_a(x)| \leq p(x)\}.$$

For any real weight w , let $M_F[i \leftarrow r]$ replace row i by $(r(e_j))_j$ and set

$$\Phi_{w,c}(F, s) = w \det M_F + \sum_{a \in J} \sum_{i=1}^n (c_a)_i \det M_F[i \leftarrow s_a].$$

An absolute maximizer means a single $Q \in \Omega_p^J$ with $|\Phi_{w,c}(Q')| \leq |\Phi_{w,c}(Q)|$ for every $Q' \in \Omega_p^J$. It need not maximize the signed polynomial.

A common inverse bound consists of a number $K > 0$ and the uniform estimate

$$p(M_F^{-1}c) \leq K \|c\|_\infty \quad (F \in \mathcal{N}_{p,\eta}, c \in V).$$

It is fixed independently of the particular frame and coefficient. When $\eta < D_p$, every such frame has nonzero determinant. Suppose $\eta < D_p$, $Q = (F, s)$ is an admissible absolute maximizer with $F \in \mathcal{N}_{p,\eta}$, and a real ρ satisfies the coefficient-detection hypothesis

$$\forall F' \in \mathcal{N}_{p,\eta} \forall x' \in V (p(x') = 1), \quad \exists a \in J, p(x' - M_{F'}^{-1}c_a) \leq K\rho.$$

Then, for any x with $p(x) = 1$, there is $a \in J$ with $1 - 2K\rho \leq |s_a(x)|$.

Assumptions

A fixed common inverse bound $K > 0$ with its uniform estimate, a fixed coefficient family, the full detection hypothesis, the positive determinant gap, admissibility, near-maximality and absolute maximality are required. There is no separate positivity assumption on ρ or w .

Conclusion

The index may depend on x , but the entire satellite family s and its maximizing configuration have already been fixed. The output evaluates the satellite having the same index as the detected coefficient.

Proof route

Detect one coefficient near the given unit vector, use its matching norming satellite, then apply the two-error estimate.

▼ Proof steps

1. **Choose the index from the supplied detection hypothesis.** Apply that hypothesis to the actual near-maximal base F and the given unit vector x . It yields one $a \in J$ and, with $z_a = M_F^{-1}c_a$,

$$p(x - z_a) \leq K\rho.$$

This is a hypothesis about all near-maximal frames, not a property inferred merely from the name of a coefficient family.

2. **Use the matching satellite, without changing the family.** For the index a just selected,

$$|s_a(y)| \leq p(y) \ (y \in V), \quad |s_a(z_a)| = p(z_a), \quad p(x - z_a) \leq K\rho.$$

The first is admissibility. The middle equality is absolute norm attainment at coefficient preimages applied to the same η, Q, c, w, a , using the determinant gap, admissibility, near-maximality and absolute maximality. The last inequality is Step 1. Together with $p(x) = 1$, they give the complete calculation

$$1 - K\rho \leq p(z_a) = |s_a(z_a)| \leq |s_a(x)| + p(z_a - x) \leq |s_a(x)| + K\rho.$$

This is the nearby norming-point estimate applied to the same functional s_a , the points x and z_a , and the error bound $K\rho$. Rearranging yields $1 - 2K\rho \leq |s_a(x)|$. No new Hahn–Banach functional replaces the already fixed s_a .

The full quantifier order is defined in coefficient detection. An actual finite net supplies it as follows. For a near-maximal F' and unit x' , contractive frame coordinates gives $\|T_{F'}x'\|_\infty \leq 1$, while the uniform inverse estimate gives $1 \leq K\|T_{F'}x'\|_\infty$. Thus membership in the coefficient annulus puts $T_{F'}x'$ in $\{c : 1/K \leq \|c\|_\infty \leq 1\}$. A center within reference distance ρ then gives, by inverse transport of the coefficient error, $p(x' - M_{F'}^{-1}c_a) \leq K\rho$. This is a close preimage from the finite net. The present theorem assumes detection explicitly and does not itself select that net.

Main citations

- absolute norm attainment at coefficient preimages —
MathlibAnnex.Satellite.absoluteMaximizer_satellite_norms_preimage
- the nearby norming-point estimate — MathlibAnnex.Satellite.abs_eval_lower_of_norms_nearby
- coefficient detection — MathlibAnnex.Satellite.CoefficientDetectsUnit
- contractive frame coordinates — MathlibAnnex.Satellite.frameCoordinates_norm_le_model
- membership in the coefficient annulus —
MathlibAnnex.Satellite.frameCoordinates_mem_coefficientAnnulus
- inverse transport of the coefficient error — MathlibAnnex.Satellite.modelDist_framePreimage_le
- a close preimage from the finite net — MathlibAnnex.Satellite.exists_net_preimage_close
- Exact declaration and proof —
MathlibAnnex.Satellite.absoluteMaximizer_satellites_almost_norm

Lean source signature (exact)

```
theorem absoluteMaximizer_satellites_almost_norm {n : ℕ}
  (M : NormModel n) {J : Type u} [Fintype J] [DecidableEq J]
  {η weight ρ : ℝ} (hηD : η < detMax M)
  (H : NearMaxInverseBound M η) (coeff : J → Coord n)
  (hdetect : CoefficientDetectsUnit M η H ρ coeff)
  {C : SatelliteConfiguration n J}
  (hC : C ∈ satelliteConfigurationSet M J)
  (hnear : C.1 ∈ nearMaxFrames M η)
  (hmax : ∀ D ∈ satelliteConfigurationSet M J,
    |configurationPolynomial weight coeff D| ≤
    |configurationPolynomial weight coeff C|)
  {x : Coord n} (hx : M.p x = 1) :
  ∃ a : J, 1 - 2 * (H.boundConstant * ρ) ≤ |C.2 a x|
```

The inverse bound and coefficient-detection property are given, not constructed by this declaration. The conclusion chooses only an index.

In the source	Mathematical meaning
Coord n	The reference space $V = \mathbb{R}^n$ with its sup norm; Coord n abbreviates $\text{Fin } n \rightarrow \mathbb{R}$. Lean uses indices $0, \dots, n-1$ and the formulas use $1, \dots, n$.
M : NormModel n; M.p	The input M contains the continuous seminorm p and constants $a_p, b_p > 0$ with $a_p\ x\ _\infty \leq p(x) \leq b_p\ x\ _\infty$ for every x . M.p x is the scalar $p(x)$.

In the source	Mathematical meaning
<code>[Fintype J] [DecidableEq J]</code>	J is finite, with equality of indices available for replacing one entry. This adds no mathematical restriction to a finite set.
<code>DualFrame n; dualFrameMatrix B; frameDet B</code>	An ordered family $F = (\ell_i)$ of continuous real linear functionals, its matrix $M_F = (\ell_i(e_j))_{i,j}$ in the standard basis, and its determinant $\det M_F$. No independence is imposed by the family type.
<code>detMax M; nearMaxFrames M η</code>	$D_p = \max_{F \in \mathcal{U}_p} \det M_F $ and $\mathcal{N}_{p,\eta} = \{F \in \mathcal{U}_p : D_p - \eta \leq \det M_F \}$, where $\mathcal{U}_p = \{(\ell_i) : \forall i, x, \ell_i(x) \leq p(x)\}$.
<code>SatelliteConfiguration n J; C.1; C.2 a</code>	A pair $Q = (F, s)$, called C . Its first component $C.1$ is the base frame F ; $C.2 \ a$ is the satellite $s_a : V \rightarrow \mathbb{R}$, continuous and real linear.
<code>weight; coeff a; coeff a i</code>	The real weight w , the coefficient vector $c_a \in V$, and its i th coordinate $(c_a)_i$. These coefficients are fixed before maximization.
<code>hC : C ∈ satelliteConfigurationSet M J</code>	The given configuration lies in Ω_p^J : $ \ell_i(x) \leq p(x)$ and $ s_a(x) \leq p(x)$ for every i, a, x .
<code>configurationPolynomial weight coeff C</code>	The scalar $\Phi_{w,c}(Q) = w \det M_F + \sum_{a \in J} \sum_i (c_a)_i \det M_F[i \leftarrow s_a]$, where the replaced row is $(s_a(e_j))_j$.
<code>hmax : ∀ D ∈ satelliteConfigurationSet M J, ...</code>	For every admissible comparison pair $Q' \in \Omega_p^J$, $ \Phi_{w,c}(Q') \leq \Phi_{w,c}(Q) $. D names that pair, not the number D_p . This is a hypothesis about absolute values.
<code>H : NearMaxInverseBound M η; H.boundConstant</code>	A fixed number $K = H.boundConstant > 0$ together with the uniform estimate $p(M_{F'}^{-1}v) \leq K\ v\ _\infty$ for every $F' \in \mathcal{N}_{p,\eta}$ and $v \in V$.
<code>hdetect : CoefficientDetectsUnit M η H ρ coeff</code>	For every $F' \in \mathcal{N}_{p,\eta}$ and x' with $p(x') = 1$, there exists $a \in J$ such that $p(x' - M_{F'}^{-1}c_a) \leq K\rho$. This is an input assumption about the fixed coefficients.

In the source	Mathematical meaning
<code>hηD; hnear; hx</code>	The three input conditions $\eta < D_p$, $F \in \mathcal{N}_{p,\eta}$ for the actual base, and $p(x) = 1$.
<code>∃ a : J, 1 - 2 * (H.boundConstant * ρ) ≤ C.2 a x </code>	There is an index a in the already fixed family such that $1 - 2K\rho \leq s_a(x) $. Only the index may depend on x .

Here $\rightarrow L[\mathbb{R}]$ means continuous real linear, $\forall$ means “for every”, and $\in$ is membership. Dot notation selects a field or component; successive arguments denote function application. Both p and weight are real, with no separate positivity hypotheses.

▼ Relevant surrounding context (separate exact excerpts)

Exact source lines 13–15:

```
private abbrev Coord (n : ℕ) := Fin n → ℝ
private abbrev NormModel (n : ℕ) := EquivalentSeminorm (Coord n)
private abbrev DualFrame (n : ℕ) := DeterminantFrame.Frame n (Coord n)
```

Exact source lines 20–21:

```
private abbrev framePreimage {n : ℕ} (B : DualFrame n) (c : Coord n) := (dualFrameMatrix B)-1.mulVec c
private abbrev IsDualContraction {n : ℕ} (M : NormModel n) (r : Coord n → L[ℝ] ℝ) := ∀ x, |r x| ≤ M.p x
```

Exact source lines 53–55:

```
private abbrev detMax {n : ℕ} (M : NormModel n) := DeterminantFrame.determinantMaximum (modelBasis M)
private abbrev NearMaxInverseBound {n : ℕ} (M : NormModel n) := DeterminantFrame.NearMaxInverseBound (modelBasis M)
private abbrev nearMaxFrames {n : ℕ} (M : NormModel n) (η : ℝ) := {B : DualFrame n | B ∈ dualFrameSet M
  ∧ detMax M - η ≤ |frameDet B|}
```

Full surrounding source. These excerpts are separate from the declaration above.

Surrounding assumptions and aliases: exact source, lines 7–55. The full original context is retained with the source evidence.

▼ Exact content identity

Declaration: MathlibAnnex.Satellite.absoluteMaximizer_satellites_almost_norm

Accepted content SHA-256: 6976bebd7ca306e9c03a6efdf537df1151073b83c843c1b9d2919aacd274f2b4

Accepted source guide SHA-256: aaeef89abc69aca6cd271cb471843acc0b569003ad32efc53027890abc4cf7a

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73